Daryl BOULOU

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EXPERIENCE

EQUANS | *AI Agent Development* — **France** | *Sept. 2025 – Jan. 2026*

- **AI Strategy & Developer Enablement**: Recruited to design an open-source AI agent using Ollama for CI/CD optimization; adapted to internal governance requirements by leading internal workshops on advanced GitHub Copilot capabilities to streamline engineering workflows.
- **Prompt Engineering for CI/CD**: Designed structured, reusable prompt templates to assist engineering teams with automated software delivery and pipeline tasks.
- **Tech Stack**: GitHub Copilot, Ollama, GitHub Actions, Prompt Engineering, Git.

AQUILAB | *QA Automation Intern* — **France** | *Jan. 2025 – Apr. 2025*

- **Test Automation & Release Validation**: Built end-to-end automated test scripts with Cypress and executed functional test cycles to validate production software releases.
- **Full-Stack Feature Development & UI/UX**: Revamped the application UI/UX for a smoother user experience and implemented a brand-new product feature to expand platform capabilities.
- **Tech Stack**: Cypress, TypeScript, JavaScript, UI/UX Design, Automated QA Testing.

EDUCATION

CESI school of engineering | *Engineering Degree in Computer Science — AI & Data Science* **France** | *2024 – Present*

Politehnica University of Timişoara (UPT) | *Academic Exchange Semester* **Timişoara, Romania** | *Mar. 2026 – Jul. 2026*

- **Relevant Coursework**: Natural Language Processing, Reinforcement Learning, Cloud Computing.

Prépa Saint-Jean | *Preparatory Classes for Engineering Schools* — **Douala, Cameroon** | *2022 – 2024*

PROJECTS

- **AI Chess Coach : Local LLM & Stockfish Analysis** GitHub
  Developed a full-stack interactive chess analysis web application combining real-time Stockfish streaming with a locally hosted LLM (`gemma4:e4b` via Ollama). Engineered a blind-first agentic tool-calling loop to ground explanations in UCI engine evaluations and eliminate LLM hallucinations. Packaged as an all-in-one multi-stage Docker container (React 19, TypeScript, FastAPI, Stockfish, Ollama).
- **Leyenda : Deep Learning Image Processing & Automated Captioning** Live Demo | GitHub
  Built and deployed a 3-stage Computer Vision & NLP pipeline: (1) a 5-class CNN classifier with ElasticNet regularization and optimal Youden thresholding ($\theta = 0.2301$) achieving 93.04% binary accuracy and 0.9758 ROC-AUC; (2) a fully convolutional U-Net image denoiser trained with combined MSE + SSIM loss; and (3) an InceptionV3 + Bahdanau Spatial Attention + LSTM decoder trained on MS-COCO achieving a 0.750 CIDEr score. Deployed via Flask, Docker, Azure Container Apps, and Cloudflare.
- **Local GitLab AI Coding Agent** GitHub
  Engineered an on-premise autonomous coding agent in Python using local LLMs and the `python-gitlab` API to monitor `ai:agent` issues, inspect repository trees, commit code fixes on dedicated branches, and automatically open Merge Requests.
- **MLP Language Model from Scratch** GitHub | Article
  Implemented Bengio et al.'s (2003) character-level Multi-Layer Perceptron language model from scratch in PyTorch using raw tensor operations, Kaiming initialization, custom Batch Normalization with exponential moving statistics, and Softmax logit scaling.

TECHNICAL SKILLS

- **Languages**: Python, TypeScript, JavaScript, SQL, Bash, HTML/CSS
- **AI / Machine Learning**: PyTorch, Keras 3, TensorFlow, Ollama (Local LLMs & Tool Calling), Computer Vision, NLP, Reinforcement Learning
- **Frameworks & Testing**: FastAPI, Flask, React, Cypress
- **Cloud & DevOps**: Docker, Azure Container Apps, Cloudflare, GitHub Actions, GitLab CI/CD, Linux
- **Spoken Languages**: French (Native), English (TOEIC: 935)